SCD LAB 2

22k-5195 Laiba Fatima

Task 1

```java
import java.util.InputMismatchException;
import java.util.Scanner;

public class q1 {
    public static void main(String[] args) {
        Scanner inp = new Scanner(System.in);
        System.out.println("Roll number:");
        String roll = inp.next();
        try {
            System.out.println("Input an integer: ");
            int num = inp.nextInt();

        } catch (InputMismatchException e) {
            System.out.println("Input value is non integer ");
        }
        System.out.println("Thanks for trying!");
    }
}
```

Run q1

```
"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDE
Roll number:
22k-5195
Input an integer:
t
Input value is non integer
Thanks for trying!

Process finished with exit code 0
```

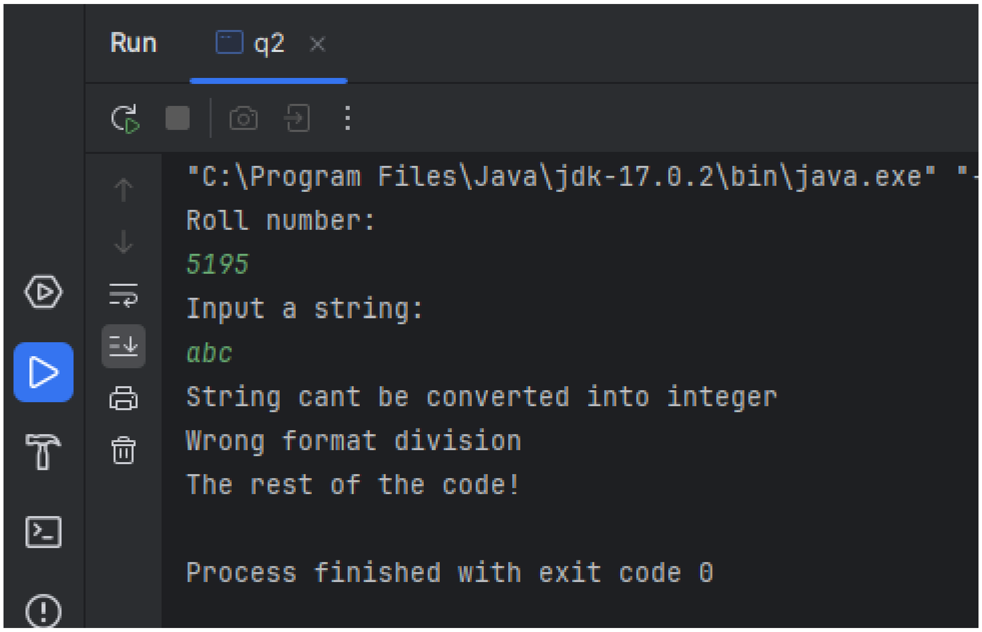
Run q1

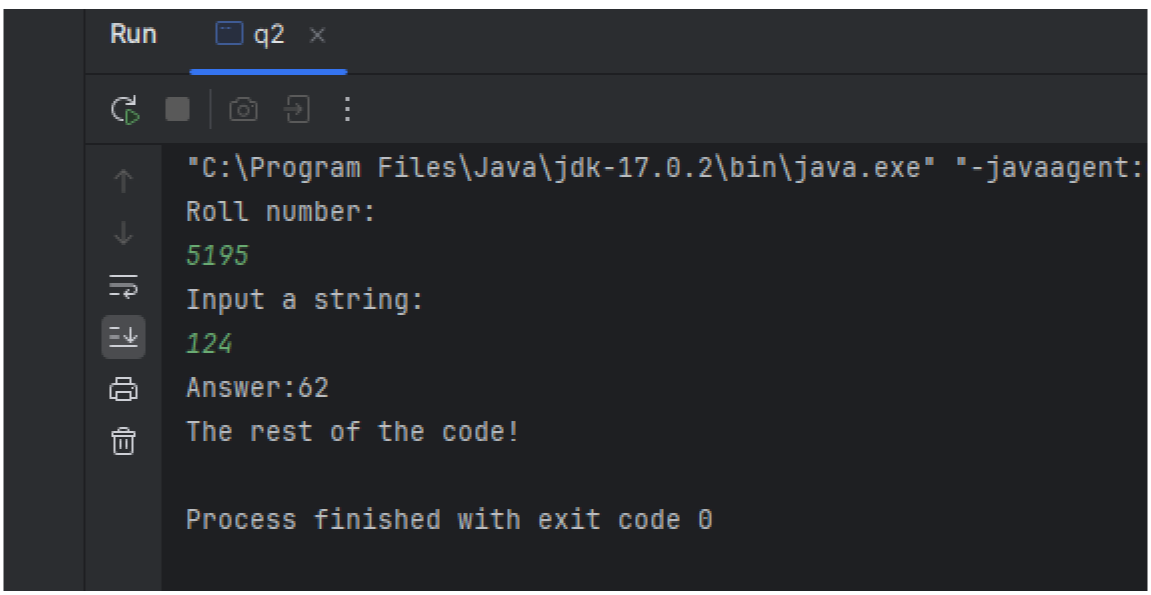
```
"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaagent:C:\Pr
Roll number:
22k-5195
Input an integer:
3
Thanks for trying!

Process finished with exit code 0
```

Task 2

```
import java.sql.SQLOutput;
import java.util.Scanner;

public class q2 {
    public static void main(String[] args) {
        Scanner inp = new Scanner(System.in);
        System.out.println("Roll number:");
        String roll = inp.next();
        int i = 0;

        try{
            try{
                System.out.println("Input a string: ");
                String num = inp.next();
                i = Integer.valueOf(num);
            }
            catch(NumberFormatException e){
                System.out.println("String cant be converted into integer");

            }

             i = i/0;
            System.out.println("Answer:" +i);

        }catch(ArithmeticException e){
            System.out.println("Wrong format division");

        }
```

```
        System.out.println("The rest of the code!");

    }
}
```





Task 3

```
import java.util.Scanner;

public class q3 {
    public static void main(String[] args) {
        Scanner inp = new Scanner(System.in);
```

```java
        System.out.println("Roll number:");
        String roll = inp.next();
        int a[] = new int[4];

        try{
            a[3] = 30/0;

        }catch(ArithmeticException e){
            System.out.println(e);
        }
        try{
            a[4] = 40/2;
        }catch(ArrayIndexOutOfBoundsException e){
            System.out.println(e);
        }

    }
}
```

```
Run    q3

"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBr
Roll number:
22k-5195
java.lang.ArithmeticException: / by zero
java.lang.ArrayIndexOutOfBoundsException: Index 4 out of bounds for length 4

Process finished with exit code 0
```

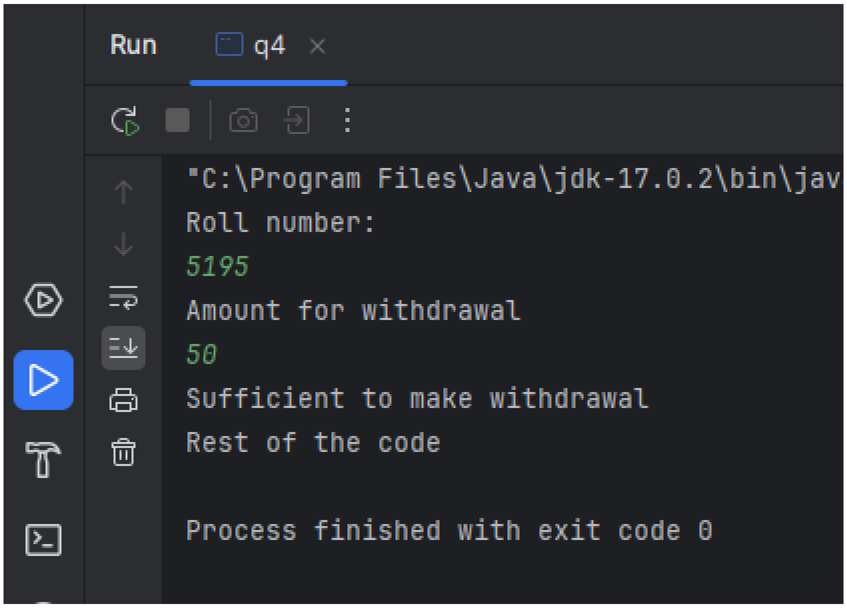
Task 4

```java
import com.sun.security.jgss.GSSUtil;

import java.util.Scanner;

class InsufficientBalanceException extends Exception{
    public InsufficientBalanceException (String str)
    {
        super(str);

    }
}
public class q4 {

    static void checkBalance(int amount) throws InsufficientBalanceException{
        int bankbalance = 170;

        if( (bankbalance - amount)<100){
```

```java
            throw new  InsufficientBalanceException("Insufficient balance");
        }else{
            System.out.println("Sufficient to make withdrawal");
        }


    }

    public static void main(String[] args) {
        Scanner inp = new Scanner(System.in);
        System.out.println("Roll number:");
        String roll = inp.next();
        System.out.println("Amount for withdrawal");
        int amount = inp.nextInt();
        try {
            checkBalance(amount);
        } catch (InsufficientBalanceException e) {
            System.out.println("Catching the custom exception");
            System.out.println(e.getMessage());
        }

        System.out.println("Rest of the code");

    }
}
```

Run   q4

```
"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaage
Roll number:
5195
Amount for withdrawal
100
Catching the custom exception
Insufficient balance
Rest of the code

Process finished with exit code 0
```



Task 5

```java
import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileReader;

public class q5 {
    public static void main(String[] args) {
        System.out.println("22k-5195");
        try{
            FileReader f =  new
FileReader("C:\\Users\\k225195\\Desktop\\laiba.txt");
            int i;
            while ((i = f.read()) != -1 ){
                System.out.println((char)i);
            }
        }catch(Exception e){

            System.out.println(e.getMessage());
        }
        finally{
            System.out.println("Closing the file");

        }
    }
}
```

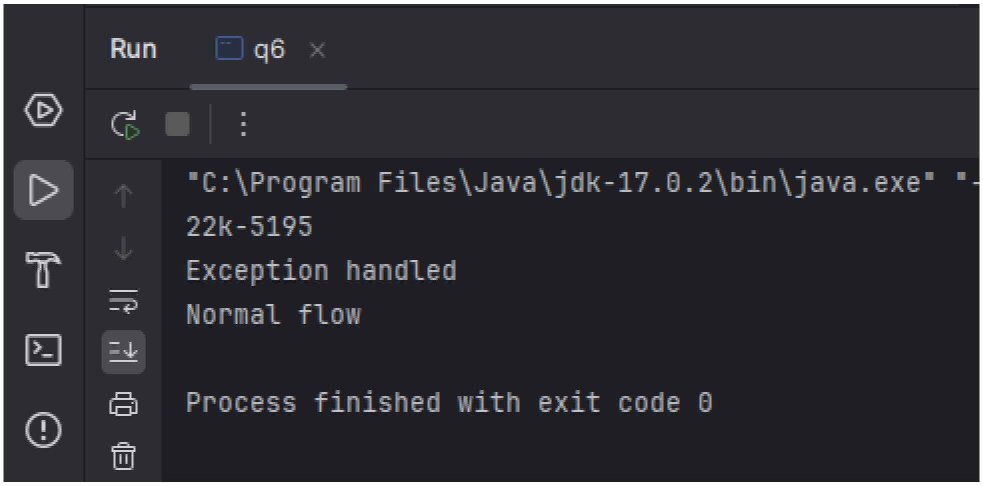
```
Run   q5

"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains
22k-5195
C:\Users\k225195\Desktop\laiba.txt (The system cannot find the file specified)
Closing the file

Process finished with exit code 0
```

```
Run   q5

"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaagent:
22k-5195
h
e
y

Closing the file

Process finished with exit code 0
```

Task 6

```java
import java.io.IOException;

 public class q6{
    void a() throws IOException {
        throw new java.io.IOException("device error");
    }
    void b() throws IOException {
        a();
    }
    void c(){
        try{
            b();
        }catch(Exception e){System.out.println("Exception handled");}
    }
    public static void main(String args[]){
        System.out.println("22k-5195");
        q6 obj=new q6();
        obj.c();
        System.out.println("Normal flow");
```

```
    }
}
```



Run
q6
"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-
22k-5195
Exception handled
Normal flow
Process finished with exit code 0